Housing Dataset: Exploratory Analysis & Predictive Modelling

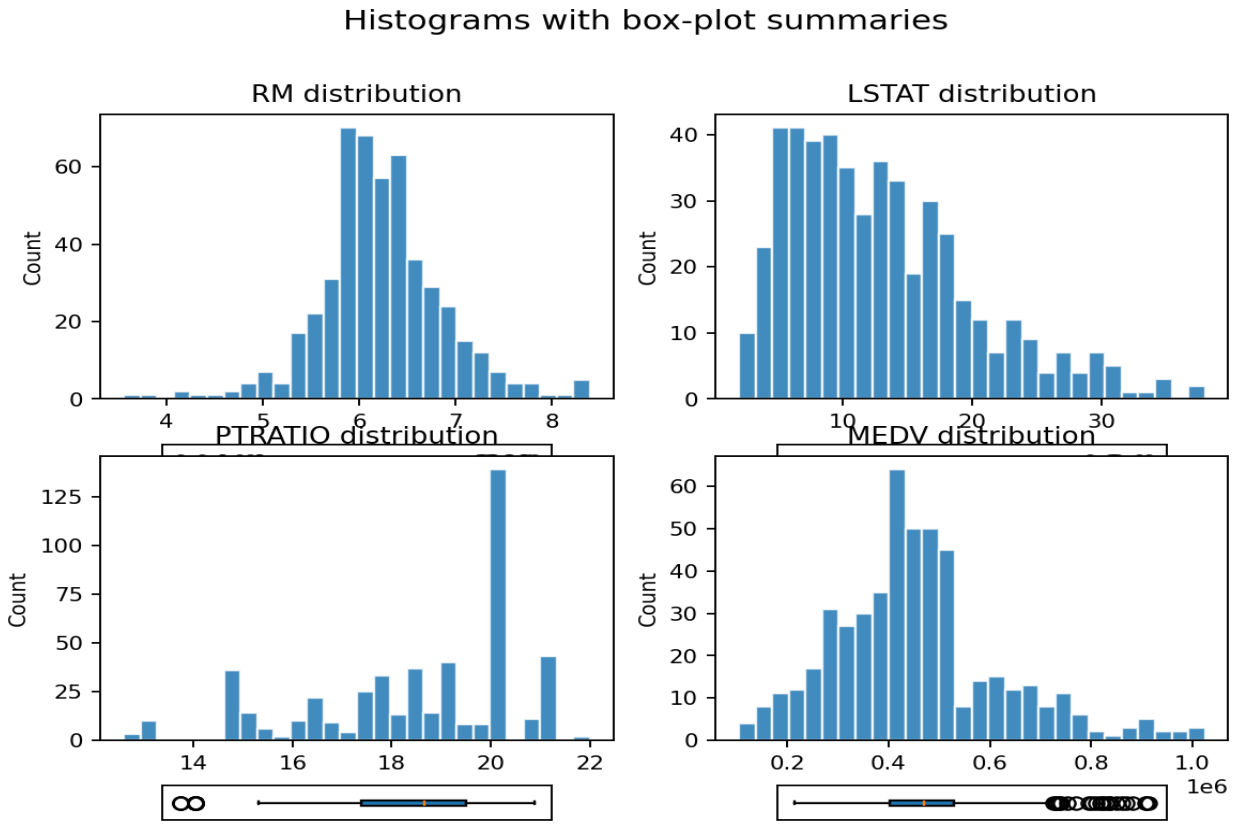
Evidence-led report generated from the supplied housing(1).csv dataset and dashboard_housing(2).py code.

Executive reading guide

Dataset size: 489 complete records and four numeric variables. The analysis code uses RM, LSTAT and PTRATIO as explanatory features and MEDV as the default prediction target. All four columns are numeric; the supplied CSV contains no missing values in these fields and no exact duplicate rows.

What this report does: describes distributions, checks unusual values, explores relationships, compares regression algorithms, examines cross-validation and model complexity, and reviews grid-search results.

Headline model result: Random Forest produced the lowest holdout RMSE in the benchmark (R^2=0.851; MAE=44,053; RMSE=57,229). This is a result on one reproducible 80/20 split, not a guarantee of performance on new data.



Important scope and measurement note. The code labels MEDV as an estimated property value and formats it with a dollar sign in some dashboard text. However, the CSV and code do not document the currency, valuation date, geography, or measurement methodology. This report therefore refers to MEDV as a recorded numeric value and does not assume a currency or location.

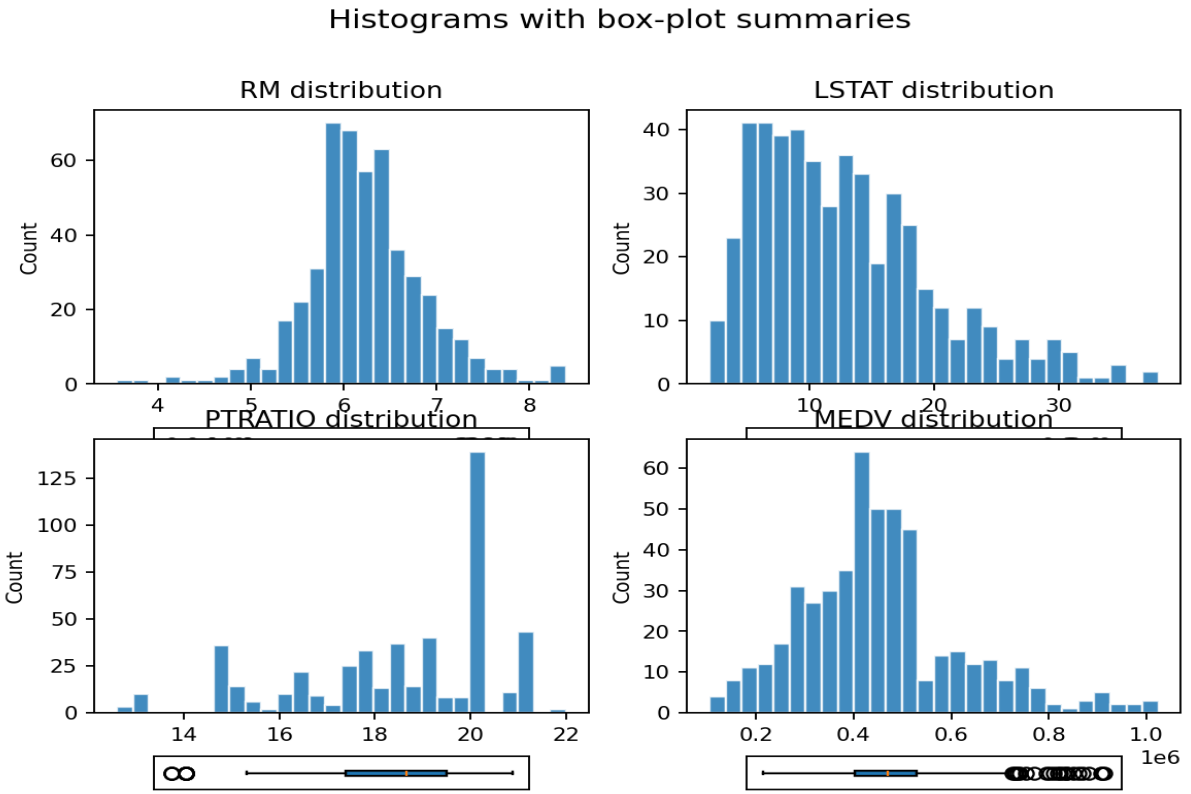
Variables, coverage and descriptive statistics

The interpretation below follows the variable definitions and limits stated in the supplied dashboard code.

Variable definitions and boundaries

RM is described in the code as the average number of rooms per dwelling. **LSTAT** is a percentage indicator supplied in the dataset; the code labels it as lower-status population percentage. **PTRATIO** is the pupil-teacher ratio. **MEDV** is the recorded median property-value target.

The four columns are the complete analytical scope. No geographic identifiers, property age, property type, or documented unit/currency field are present. Do not infer those missing attributes from the numeric values.



Descriptive statistics

Variable	Count	Mean	Std. dev.	Minimum	Median	Maximum	Skewness
RM	489	6.24	0.64	3.56	6.18	8.40	0.08
LSTAT	489	12.94	7.08	1.98	11.69	37.97	0.91
PTRATIO	489	18.52	2.11	12.60	19.10	22.00	-0.82
MEDV	489	454,342.94	165,340.28	105,000.00	438,900.00	1,024,800.00	0.78

Skewness is a compact indicator of distribution asymmetry: values above zero indicate a longer right tail, values below zero a longer left tail. It should be read together with the histogram and box plot rather than in isolation.

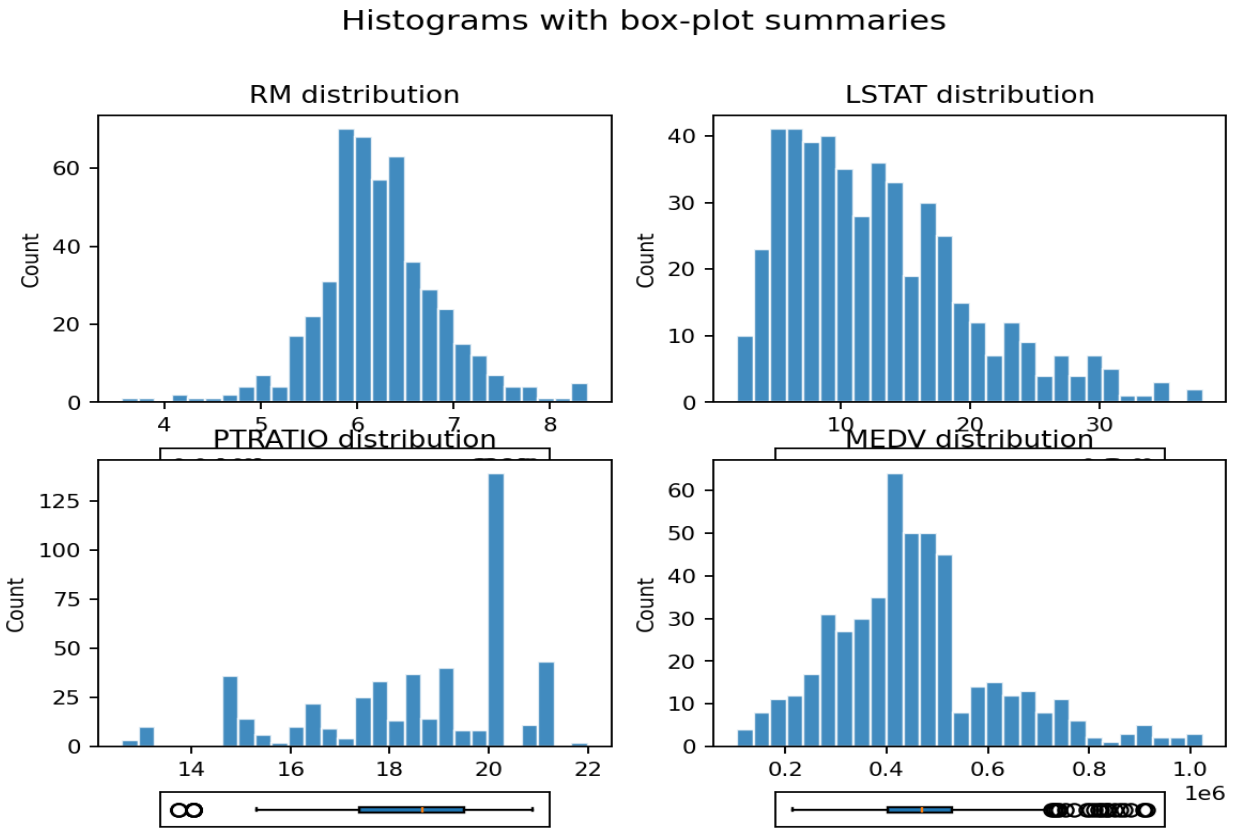
Distributions and potential outliers

Histograms show frequency patterns; box plots summarize the median, quartiles and potential extreme observations.

How to read the distributions

Read each histogram for concentration, spread, skew and possible multiple peaks. The box plot below each histogram shows the median and interquartile range (IQR). The dashboard uses the conventional 1.5 × IQR rule to flag values outside the lower and upper fences. Such observations are **potential outliers**, not automatic errors.

Investigate flagged records in context before excluding them. A valid but unusual property record can carry important information. Also check whether the distribution suggests a transformation or robust modelling approach.



IQR screening results

Variable	Q1	Q3	IQR lower fence	IQR upper fence	Flagged rows
RM	5.88	6.58	4.84	7.62	22
LSTAT	7.37	17.12	-7.25	31.75	7
PTRATIO	17.40	20.20	13.20	24.40	13

Variable	Q1	Q3	IQR lower fence	IQR upper fence	Flagged rows
MEDV	350,700.00	518,700.00	98,700.00	770,700.00	22

Relationships with MEDV and correlation

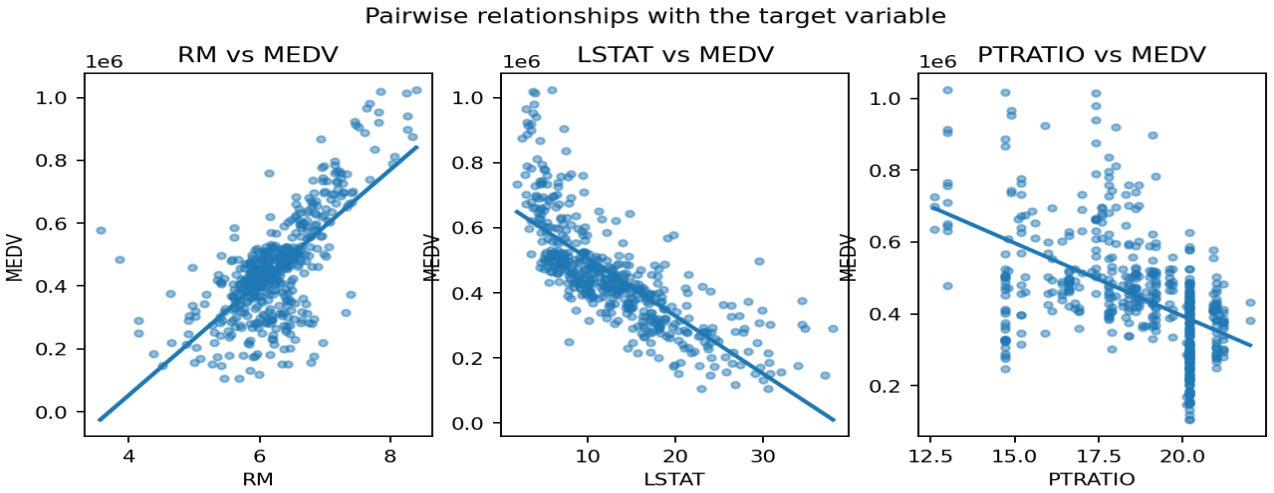
Scatter plots show the observed pairs; the straight line is a simple least-squares trend, not a causal model.

Feature-by-feature interpretation

Each panel compares one explanatory variable with MEDV. The slope indicates the direction of the sample's linear trend; the dispersion around the line shows how much variation remains unexplained by that single feature.

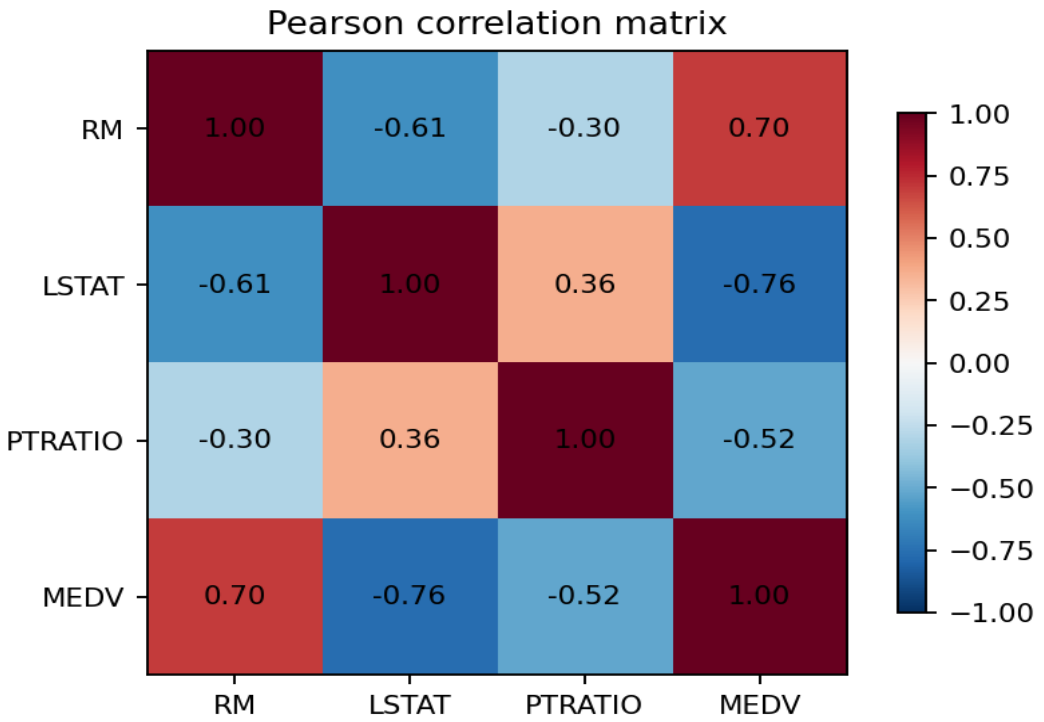
RM vs MEDV: positive, moderate Pearson association (r=0.697).
LSTAT vs MEDV: negative, strong Pearson association (r=-0.761).
PTRATIO vs MEDV: negative, moderate Pearson association (r=-0.519).

Pearson correlation measures linear association only. It does not establish cause and effect, and a near-zero correlation does not rule out a nonlinear relationship.



Correlation across all four variables

The heatmap reports Pearson correlation coefficients from −1 to +1. Values near +1 indicate strong positive linear association; values near −1 indicate strong negative association; values near zero indicate weak linear association. Use the matrix to identify overlapping information among predictors and to guide further diagnostics.



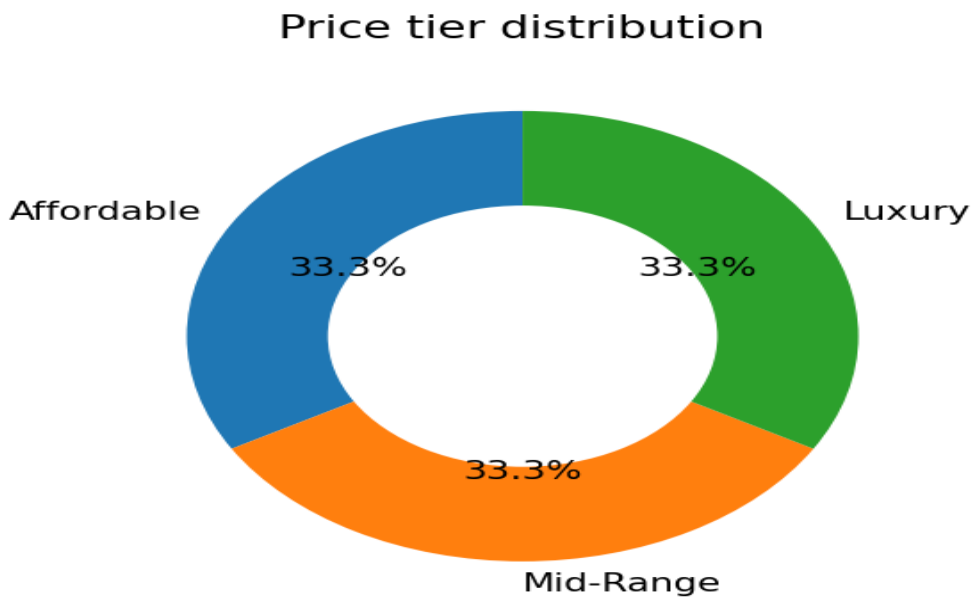
Price tiers, averages and multivariable views

Price tiers are created by dividing MEDV ranks into three approximately equal groups.

Tier construction and interpretation

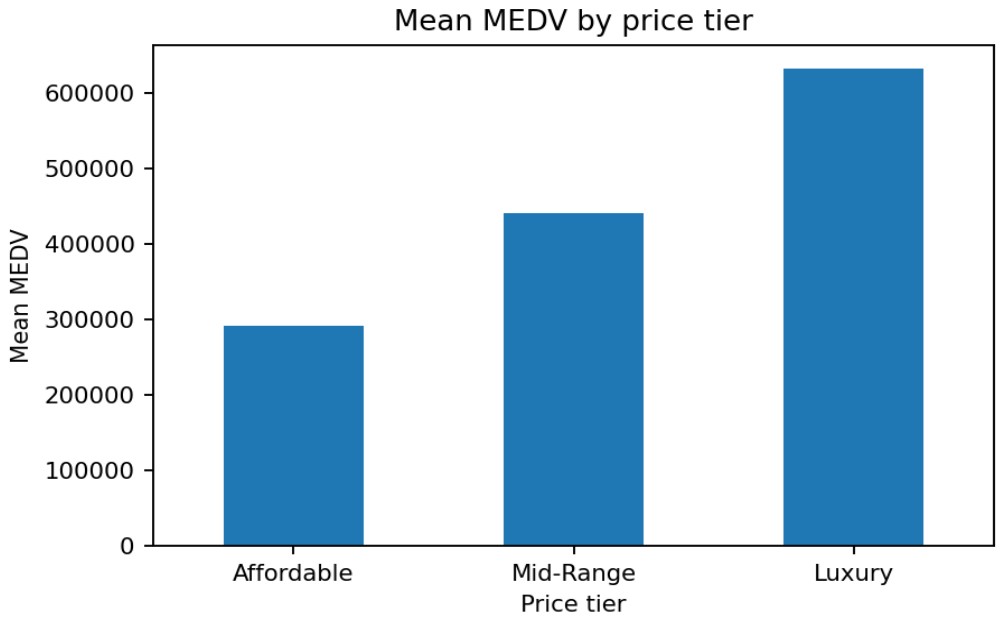
The code creates Affordable, Mid-Range and Luxury categories using qcut on the rank of MEDV, with one-third of records assigned to each tier. These are relative groups within this dataset, not externally defined market segments.

The pie chart shows the number of records in each group. The bar chart compares average MEDV by tier. Because the tiers are derived directly from MEDV, separation in average MEDV is expected by construction and should not be treated as independent validation of a market segmentation.



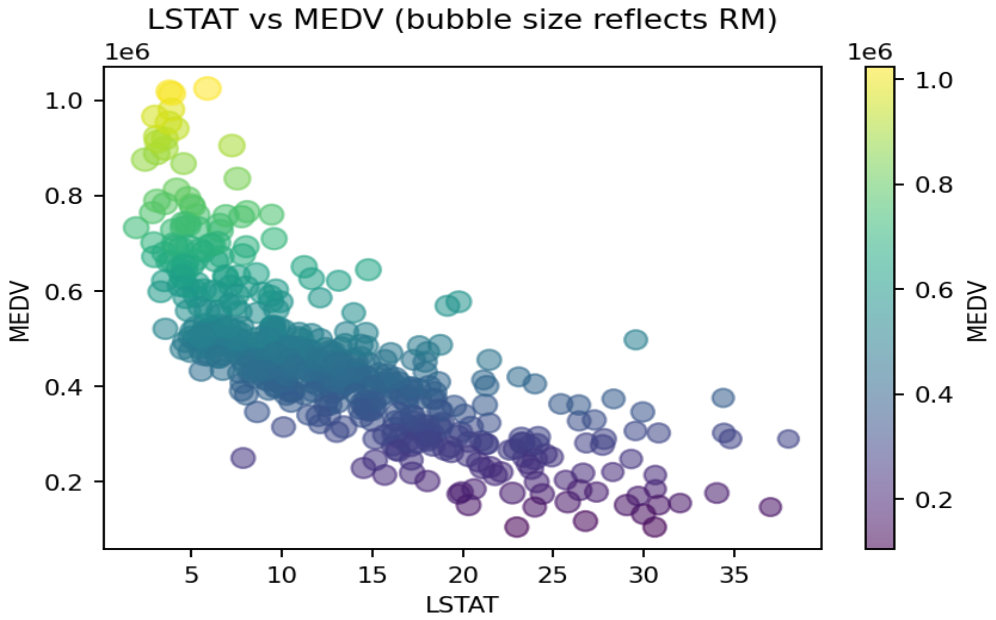
Mean MEDV by tier

This bar chart summarizes the mean target value in each rank-based group. It is useful as a descriptive comparison, but it does not reveal the full spread within each tier. Combine it with the distribution and box-plot views before describing differences between groups.



Bubble scatter: LSTAT and MEDV

The dashboard's bubble plot uses LSTAT on the horizontal axis, MEDV on the vertical axis, and RM to size the markers. It adds a third variable to a two-dimensional scatter view. Overlapping bubbles can obscure records, so use the non-sized scatter plots for a clearer view of point density.



Linear versus polynomial regression

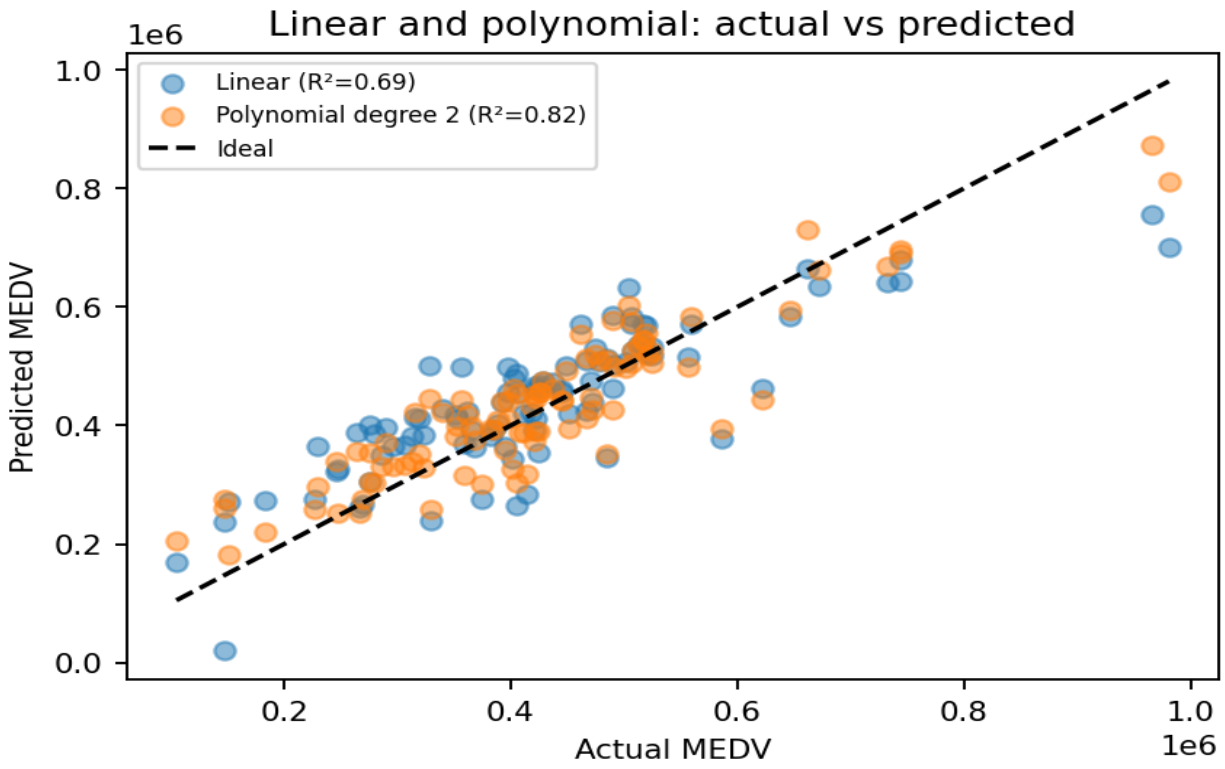
Both models use the same 80/20 holdout split (random_state=42) and three predictors: RM, LSTAT and PTRATIO.

Actual versus predicted values

Linear regression: R^2=0.691. **Polynomial regression, degree 2:** R^2=0.818.

The polynomial pipeline expands the predictors to include degree-two terms before fitting a linear estimator. The diagonal is the ideal prediction line: points close to it have smaller prediction errors.

On this holdout split, the degree-two model achieves a higher R^2 than the simple linear model. This indicates a better fit to this test subset, but it should be checked with cross-validation and a separate evaluation sample before being adopted.



Interpretation of evaluation metrics

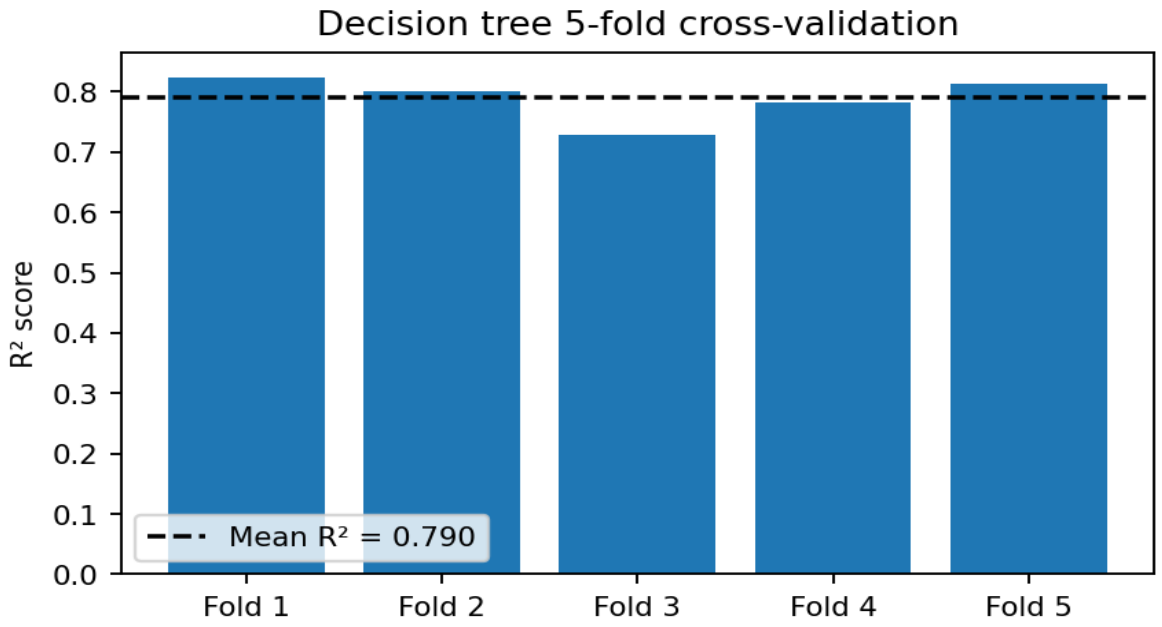
Metric	Meaning	Preferred direction
R^2	Proportion of target variance accounted for by the model on the evaluation sample.	Higher is generally better; can be negative.
MAE	Average absolute prediction error in MEDV units.	Lower is better; easier to interpret in target units.
RMSE	Square root of mean squared error; gives greater weight to large errors.	Lower is better; sensitive to large errors.

Cross-validation, model complexity and overfitting

Validation helps estimate stability and diagnose the trade-off between model flexibility and generalization.

K-fold cross-validation

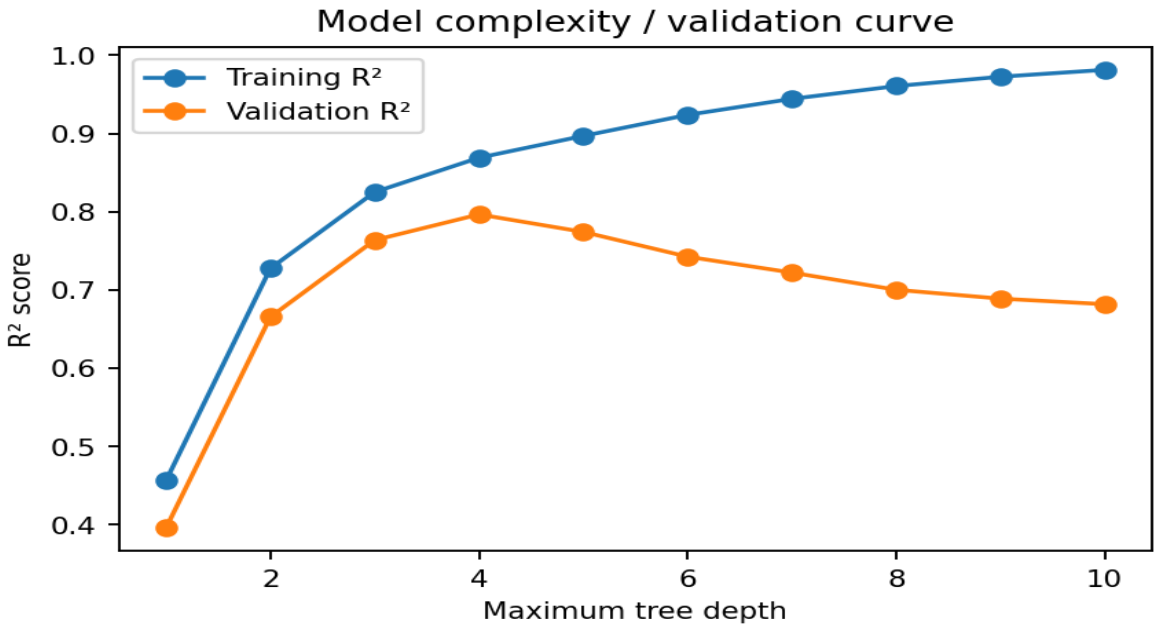
The dashboard evaluates a depth-5 decision tree with 5-fold shuffled K-Fold validation. The mean R^2 is 0.790 and the standard deviation across folds is 0.033. Each bar represents one fold. Similar fold scores suggest more consistent performance across the partitions; large variation suggests sensitivity to which observations are held out.



Model complexity and fit diagnostics

The validation curve changes the decision tree's maximum depth from 1 to 10 and averages training and validation R^2 over repeated ShuffleSplit partitions. Training performance typically increases with complexity. Validation performance may improve and then flatten or decline. A widening gap between the two curves is a warning sign for overfitting.

The dashboard flags a large final training-validation gap (greater than 0.15) as high variance and low scores on both curves as high bias. These are practical heuristics, not universal statistical thresholds.



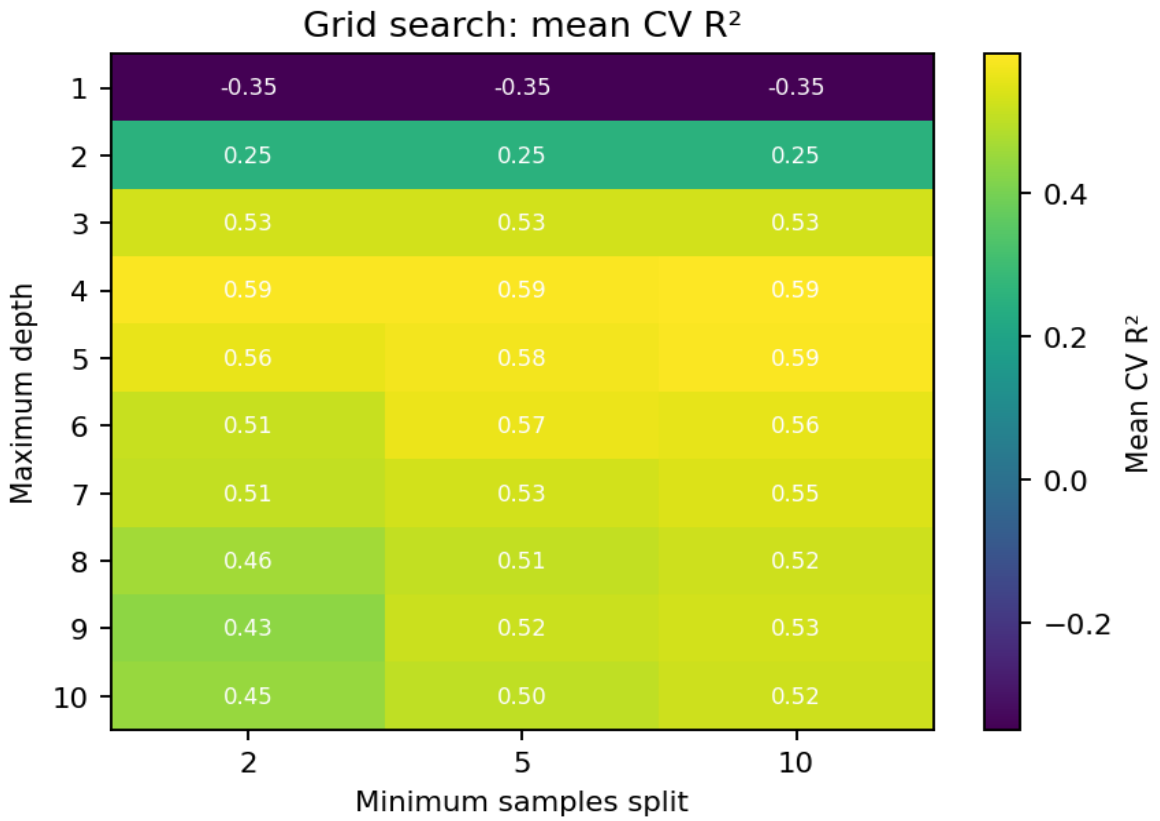
Grid search: selecting decision-tree settings

The heatmap summarizes mean 5-fold cross-validation R^2 for combinations of tree depth and minimum split size.

Grid-search heatmap and selected parameters

The dashboard searches maximum depth values from 1 to 10 and minimum-sample-split values of 2, 5 and 10. The best combination in this run is **max_depth=4** and **min_samples_split=10**, with mean CV R^2=0.594. A total of 30 parameter combinations were evaluated.

The best score is an estimate from the tuning folds, not an unbiased final test score. For a rigorous final estimate, keep a separate test set untouched during tuning or use nested cross-validation. Also consider whether the selected parameter range is broad enough for the problem.



How to use the heatmap

Compare cells within the same color scale. A higher mean CV R^2 indicates a better average score for that parameter pair. If several combinations perform similarly, prefer a simpler tree (for example, shallower depth) when it improves interpretability and reduces overfitting risk.

Eight regression models compared on a common holdout set

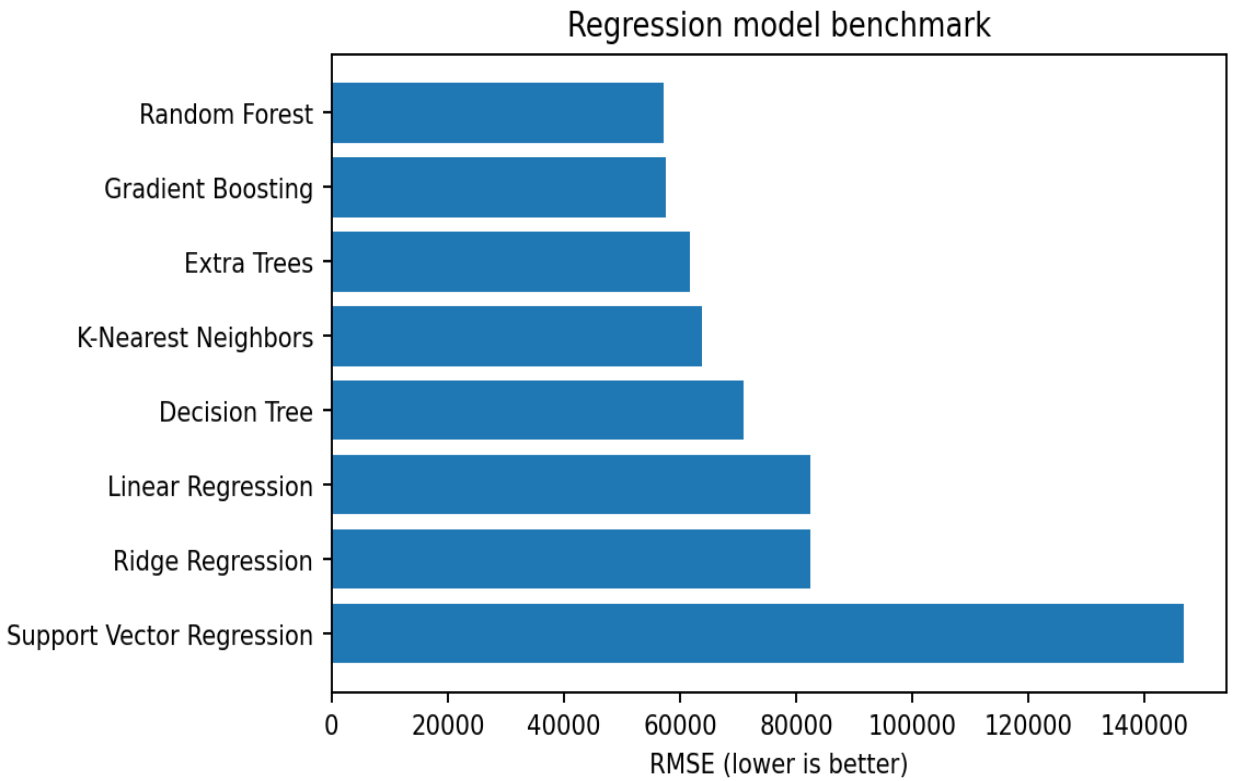
All algorithms use the same 80/20 train-test split and random_state=42 for comparability.

Holdout RMSE comparison

Best by holdout RMSE: Random Forest. Its R^2 is 0.851, MAE is 44,053, and RMSE is 57,229.

The chart orders models by RMSE; shorter bars indicate smaller typical squared-error magnitude. Gradient Boosting is close to Random Forest in this run. The gap between the best and worst models is substantial, showing that algorithm choice matters for this dataset and split.

This comparison is a benchmark, not proof that the top model will remain best under a different sample, new data, or revised preprocessing.



Model performance table (sorted by RMSE)

Model	R^2	MAE	RMSE
Random Forest	0.851	44,053	57,229
Gradient Boosting	0.849	44,113	57,551
Extra Trees	0.827	47,241	61,685
K-Nearest Neighbors	0.815	48,763	63,810

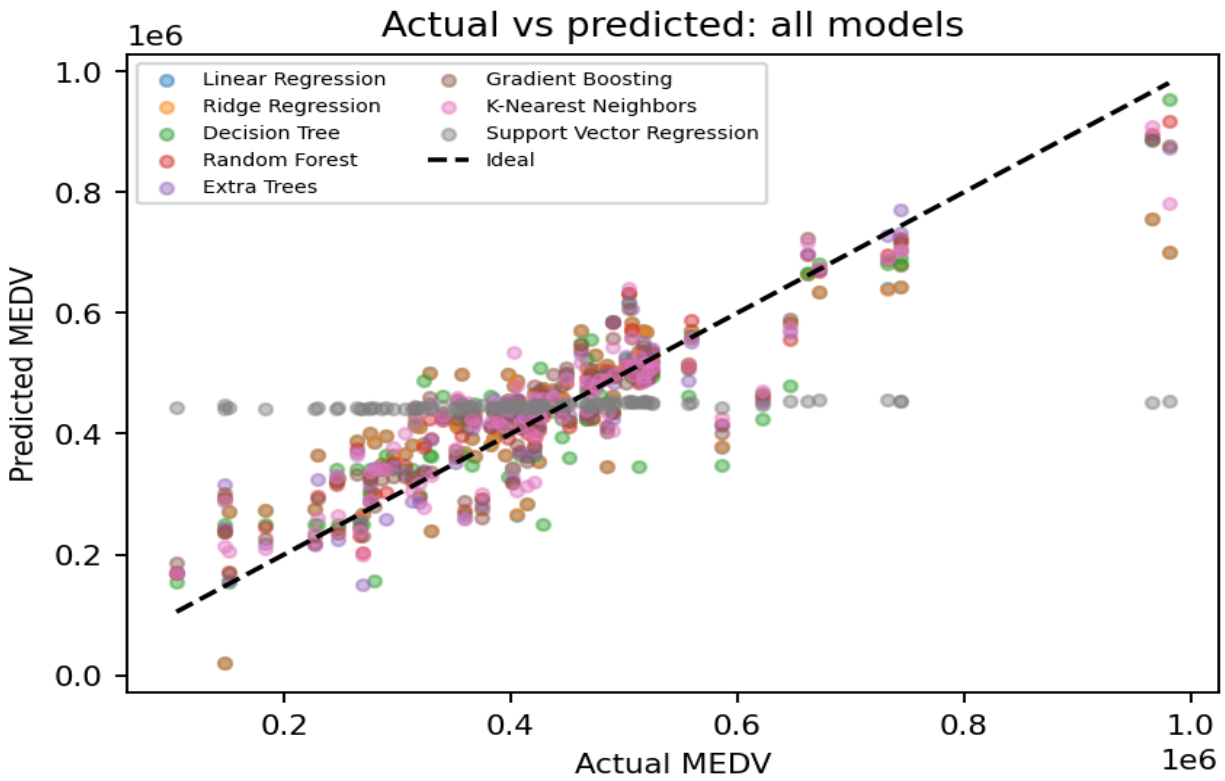
Model	R^2	MAE	RMSE
Decision Tree	0.770	53,041	71,031
Linear Regression	0.691	64,277	82,396
Ridge Regression	0.691	64,276	82,413
Support Vector Regression	0.020	106,379	146,751

Predicted values and feature contribution

Prediction plots help reveal systematic bias, dispersion and the relative role of input features.

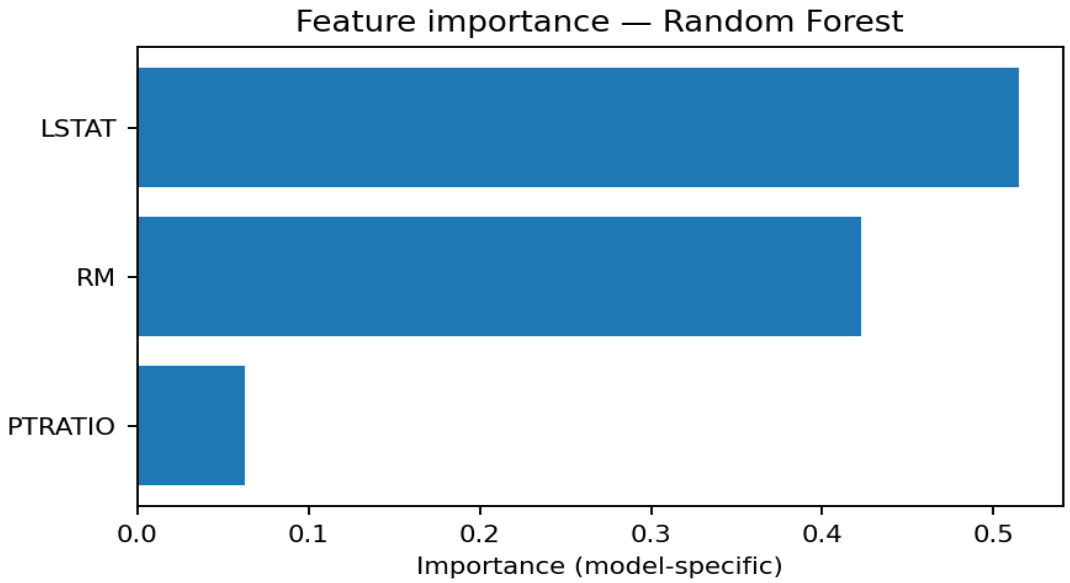
Actual versus predicted: all models

Each cloud compares test-set actual values with predictions from one of the eight algorithms. The dashed diagonal represents perfect agreement. Points consistently above or below the line indicate over- or under-prediction in that range. Dense clusters and outliers can be difficult to distinguish when all models are overlaid, so inspect models individually when making operational decisions.



Feature importance / absolute coefficients

The selected model is Random Forest. The chart displays its built-in feature importance if it is a tree model, or absolute model coefficients when exposed by the estimator. These values are model-specific and may not be directly comparable across algorithms. Importance indicates predictive contribution within the fitted model; it does not establish causal influence.



What the analysis supports — and what it does not

A cautious interpretation keeps descriptive patterns, predictive results and causal claims separate.

Finding	Interpretation
489 complete rows; 4 numeric fields.	The analytical dataset is compact and has no missing values or exact duplicates in the four required fields.
Best holdout model: Random Forest (R^2=0.851).	Random Forest is the strongest of the tested models on the specified 80/20 split, using RMSE as the ranking metric.
Polynomial R^2=0.818 vs linear R^2=0.691.	Degree-two terms improved the holdout fit relative to the simple linear model in this split.
Grid-search best CV R^2=0.594.	The selected tree parameters are the best among the tested combinations, not necessarily globally optimal.
Correlations and feature importance are associational.	Neither measure proves that changing a predictor will cause MEDV to change.

Recommended analytical workflow

1. **Validate definitions:** document the meaning, unit, currency, date and provenance of every variable, especially MEDV and LSTAT.
2. **Inspect records:** review IQR flags and extreme values against source records; do not delete them automatically.
3. **Establish a baseline:** retain linear regression as an interpretable reference and compare it with nonlinear models.
4. **Validate robustly:** use repeated or nested cross-validation and preserve a final untouched test set for unbiased reporting.
5. **Improve predictors:** only add variables when their definitions, quality and availability are documented. The current file has no location, age or property-type fields.
6. **Communicate uncertainty:** report MAE/RMSE in the target's documented units, explain the evaluation split, and avoid describing estimates as formal property valuations.

Reproducibility note: the supplied dashboard uses a fixed random_state=42 for the main train-test split and several model components. Results in this report are calculated directly from the supplied CSV using the algorithms and principal settings visible in the supplied Python code. Small differences may occur across software versions.

Source files: dashboard_housing(2).py and housing(1).csv supplied in this conversation. No external data or external factual sources were added.